AERIS Expert Review Packet

Anomaly

Source / metric	tceq / no2
Observed / expected	9.7 / 1.2292
Time	2026-07-11 20:00 UTC
Location	29.5831, -95.0155
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	humidity	percent	6 / 657	46.79 to 100; mean 79.2484	70.35 at 2026-07-11 19:55Z; dt 5 min
asos	temperature	degC	6 / 657	23.6111 to 34; mean 28.1131	30 at 2026-07-11 19:55Z; dt 5 min
asos	wind_direction	degree	6 / 766	0 to 360; mean 105.065	20 at 2026-07-11 19:55Z; dt 5 min
asos	wind_speed	m/s	6 / 778	0 to 9.25999; mean 2.0862	4.11555 at 2026-07-11 19:55Z; dt 5 min
noaa_gfs	gh_500	m	6 / 72	5905.85 to 5937.07; mean 5925.31	5930.86 at 2026-07-11 18:00Z; dt 120 min
noaa_gfs	pbl_height	m	6 / 72	11.5161 to 1969.58; mean 611.878	317.356 at 2026-07-11 18:00Z; dt 120 min
noaa_gfs	precipitable_water	mm	6 / 72	51.6459 to 58.1093; mean 55.0886	56.1325 at 2026-07-11 18:00Z; dt 120 min
noaa_gfs	surface_pressure	hPa	6 / 72	1012.15 to 1017.74; mean 1015.09	1017.46 at 2026-07-11 18:00Z; dt 120 min
noaa_gfs	t_850	degC	6 / 72	17.2076 to 19.6019; mean 18.714	18.7107 at 2026-07-11 18:00Z; dt 120 min
noaa_gfs	u_10m	m/s	6 / 72	-3.47027 to 0.762437; mean -1.2036	0.762437 at 2026-07-11 18:00Z; dt 120 min
noaa_gfs	v_10m	m/s	6 / 72	-0.626672 to 4.76385; mean 2.1496	1.37225 at 2026-07-11 18:00Z; dt 120 min
openaq	ozone	ppm	15 / 989	-0.004 to 0.064; mean 0.0152	0.033 at 2026-07-11 20:00Z; dt 0 min
openaq	pm10	ug/m3	2 / 139	10 to 68; mean 29.9209	23 at 2026-07-11 20:00Z; dt 0 min
openaq	pm25	ug/m3	66 / 3490	0 to 42.1; mean 5.6328	2.2 at 2026-07-11 20:00Z; dt 0 min
openweather	cloud_cover	percent	3 / 215	3 to 100; mean 66.7302	98 at 2026-07-11 20:01Z; dt 1.9 min
openweather	humidity	percent	3 / 215	58 to 96; mean 81.6698	74 at 2026-07-11 20:01Z; dt 1.9 min
openweather	precipitation	mm/h	3 / 215	0 to 4.49; mean 0.1185	0 at 2026-07-11 20:01Z; dt 1.9 min
openweather	pressure	hPa	3 / 215	1014 to 1019; mean 1016.61	1017 at 2026-07-11 20:01Z; dt 1.9 min
openweather	temperature	degC	3 / 215	25.34 to 34.47; mean 28.5613	31.15 at 2026-07-11 20:01Z; dt 1.9 min
openweather	wind_direction	degree	3 / 215	0 to 252; mean 158.921	164 at 2026-07-11 20:01Z; dt 1.9 min
openweather	wind_speed	m/s	3 / 215	0.45 to 7.3; mean 2.6681	3.44 at 2026-07-11 20:01Z; dt 1.9 min
purpleair	pm25	ug/m3	62 / 4241	0 to 59.2; mean 7.2641	3.3 at 2026-07-11 19:59Z; dt 0 min

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
sentinel5p	s5p_aer_ai_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-07-11 19:19Z; dt 40.9 min
sentinel5p	s5p_ch4_granule_available	count	5 / 5	1 to 1; mean 1	1 at 2026-07-11 19:19Z; dt 40.9 min
sentinel5p	s5p_co_column	mol/m^2	4 / 4	0.0250428 to 0.0262775; mean 0.0256	0.0250428 at 2026-07-12 18:33Z; dt 1353.3 min
sentinel5p	s5p_co_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-07-11 19:19Z; dt 40.9 min
sentinel5p	s5p_hcho_column	mol/m^2	4 / 4	0.000154242 to 0.000254783; mean 0.0002	0.000253386 at 2026-07-12 18:33Z; dt 1353.3 min
sentinel5p	s5p_hcho_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-07-11 19:19Z; dt 40.9 min
sentinel5p	s5p_no2_column	mol/m^2	2 / 2	3.3326e-05 to 4.40738e-05; mean 0	3.3326e-05 at 2026-07-12 19:04Z; dt 1384.1 min
sentinel5p	s5p_no2_granule_available	count	4 / 4	1 to 1; mean 1	1 at 2026-07-11 19:19Z; dt 40.9 min
sentinel5p	s5p_o3_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-07-11 19:19Z; dt 40.9 min
sentinel5p	s5p_so2_column	mol/m^2	4 / 4	-5.8828e-06 to 4.71798e-05; mean 0	4.71798e-05 at 2026-07-12 18:33Z; dt 1353.3 min
sentinel5p	s5p_so2_granule_available	count	7 / 7	1 to 1; mean 1	1 at 2026-07-11 19:19Z; dt 40.9 min
tceq	co	ppm	3 / 191	0.1 to 0.9; mean 0.2482	0.2 at 2026-07-11 20:00Z; dt 0 min
tceq	no2	ppb	11 / 675	-2.7 to 19.9; mean 5.0461	9.7 at 2026-07-11 20:00Z; dt 0 min
tceq	so2	ppb	3 / 188	-0.6 to 3.7; mean -0.1729	0.7 at 2026-07-11 20:00Z; dt 0 min

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

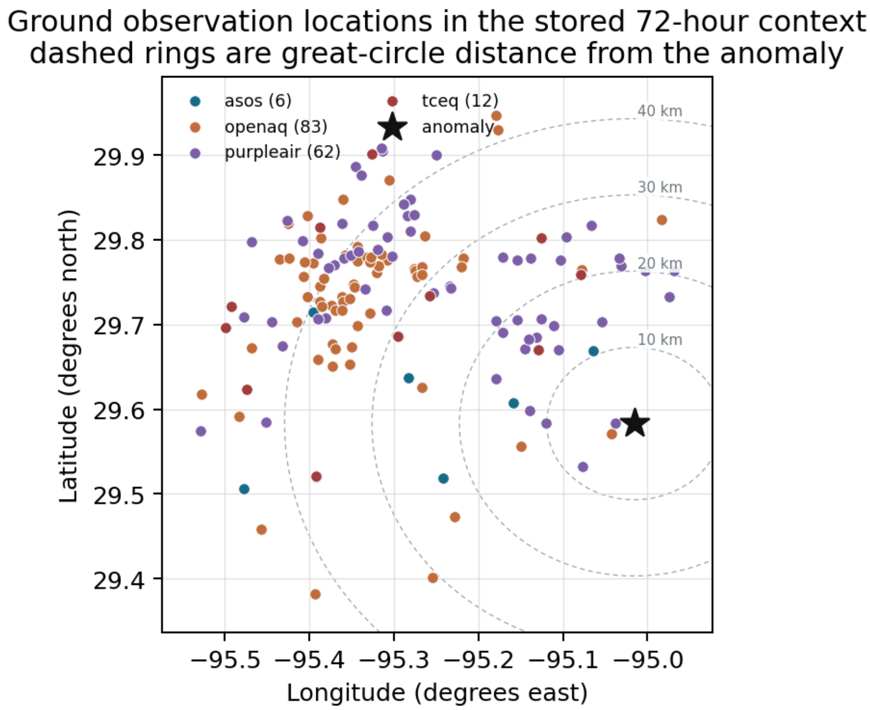
Source	Metric	Values
asos	humidity	07-10 06Z 87, 12Z 74.93, 18Z 62.61, 07-11 00Z 78.33, 06Z 88.14, 12Z 79.51, 18Z 70.31, 07-12 00Z 86.42, 06Z 87.1, 12Z 68.94, 18Z 81.73, 07-13 00Z 87.89, 06Z 89.36
asos	temperature	07-10 06Z 26.2, 12Z 29.06, 18Z 31.29, 07-11 00Z 27.74, 06Z 26.16, 12Z 27.9, 18Z 29.48, 07-12 00Z 27.11, 06Z 26.58, 12Z 30.45, 18Z 27.45, 07-13 00Z 27.26, 06Z 26.76
asos	wind_direction	07-10 06Z 55.71, 12Z 114.5, 18Z 159.2, 07-11 00Z 129.1, 06Z 58.41, 12Z 135.3, 18Z 108.7, 07-12 00Z 88.1, 06Z 58, 12Z 83.33, 18Z 149.1, 07-13 00Z 99.69, 06Z 120.9
asos	wind_speed	07-10 06Z 0.7483, 12Z 2.224, 18Z 4.408, 07-11 00Z 2.259, 06Z 0.8922, 12Z 2.24, 18Z 2.645, 07-12 00Z 2.041, 06Z 1.045, 12Z 1.403, 18Z 2.985, 07-13 00Z 1.82, 06Z 1.73
noaa_gfs	gh_500	07-10 12Z 5907, 18Z 5920, 07-11 00Z 5919, 06Z 5924, 12Z 5919, 18Z 5931, 07-12 00Z 5927, 06Z 5933, 12Z 5925, 18Z 5933, 07-13 00Z 5928, 06Z 5937
noaa_gfs	pbl_height	07-10 12Z 123.4, 18Z 1535, 07-11 00Z 851.4, 06Z 237.7, 12Z 79.07, 18Z 813.1, 07-12 00Z 691.6, 06Z 183.2, 12Z 67.81, 18Z 1777, 07-13 00Z 734.2, 06Z 249.8
noaa_gfs	precipitable_water	07-10 12Z 53.77, 18Z 55.9, 07-11 00Z 53.54, 06Z 53.08, 12Z 56.2, 18Z 55.95, 07-12 00Z 56.29, 06Z 56.98, 12Z 52.37, 18Z 54.59, 07-13 00Z 56.41, 06Z 55.98
noaa_gfs	surface_pressure	07-10 12Z 1014, 18Z 1015, 07-11 00Z 1013, 06Z 1014, 12Z 1015, 18Z 1016, 07-12 00Z 1014, 06Z 1015, 12Z 1016, 18Z 1016, 07-13 00Z 1015, 06Z 1016
noaa_gfs	t_850	07-10 12Z 18.59, 18Z 17.58, 07-11 00Z 18.66, 06Z 19.38, 12Z 18.77, 18Z 18.36, 07-12 00Z 19.28, 06Z 19.34, 12Z 18.78, 18Z 17.74, 07-13 00Z 19.07, 06Z 19.02
noaa_gfs	u_10m	07-10 12Z 0.04479, 18Z -2.298, 07-11 00Z -1.742, 06Z -1.193, 12Z -0.4219, 18Z -0.1576, 07-12 00Z -2.637, 06Z -1.213, 12Z -1.007, 18Z -2.371, 07-13 00Z -1.489, 06Z 0.0417

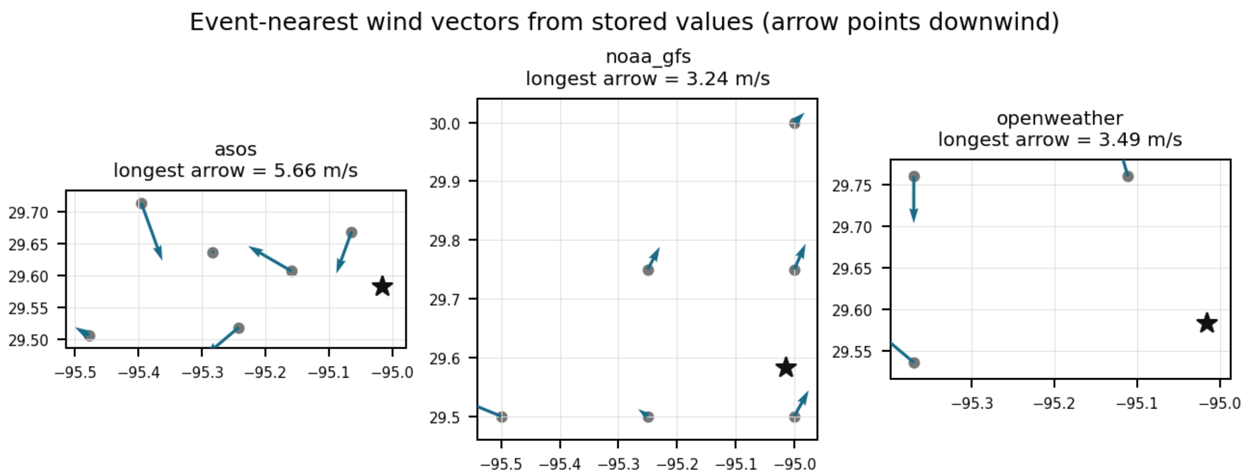
Source	Metric	Values
noaa_gfs	v_10m	07-10 12Z 1.44, 18Z 3.522, 07-11 00Z 3.608, 06Z 2.276, 12Z 0.6067, 18Z 1.007, 07-12 00Z 2.118, 06Z 1.946, 12Z 0.4241, 18Z 1.906, 07-13 00Z 4.247, 06Z 2.693
openaq	ozone	07-10 06Z 0.00675, 12Z 0.01152, 18Z 0.0224, 07-11 00Z 0.01146, 06Z 0.005658, 12Z 0.0122, 18Z 0.02844, 07-12 00Z 0.0186, 06Z 0.008, 12Z 0.01316, 18Z 0.02892, 07-13 00Z 0.01179, 06Z 0.01045
openaq	pm10	07-10 06Z 18.88, 12Z 28.33, 18Z 35.27, 07-11 00Z 42.45, 06Z 33.08, 12Z 34.27, 18Z 23.17, 07-12 00Z 23.73, 06Z 24.25, 12Z 48.92, 18Z 28.67, 07-13 00Z 17.92, 06Z 24.67
openaq	pm25	07-10 06Z 5.804, 12Z 6.381, 18Z 5.097, 07-11 00Z 6.105, 06Z 6.16, 12Z 5.405, 18Z 5.537, 07-12 00Z 5.622, 06Z 5.821, 12Z 8.229, 18Z 4.762, 07-13 00Z 3.792, 06Z 3.763
openweather	cloud_cover	07-10 06Z 45.17, 12Z 86.11, 18Z 89.59, 07-11 00Z 26.68, 06Z 12.18, 12Z 78.47, 18Z 94.53, 07-12 00Z 28.32, 06Z 70.47, 12Z 98.5, 18Z 77.94, 07-13 00Z 80.79, 06Z 94.6
openweather	humidity	07-10 06Z 88.92, 12Z 79.94, 18Z 68, 07-11 00Z 77.21, 06Z 87.76, 12Z 87.21, 18Z 73.76, 07-12 00Z 87.37, 06Z 89.71, 12Z 79.56, 18Z 71, 07-13 00Z 88.68, 06Z 89.4
openweather	precipitation	07-10 06Z 0, 12Z 0.1856, 18Z 0.005882, 07-11 00Z 0, 06Z 0, 12Z 0.5068, 18Z 0.2094, 07-12 00Z 0.1079, 06Z 0, 12Z 0, 18Z 0.3772, 07-13 00Z 0, 06Z 0
openweather	pressure	07-10 06Z 1015, 12Z 1017, 18Z 1015, 07-11 00Z 1015, 06Z 1016, 12Z 1018, 18Z 1016, 07-12 00Z 1016, 06Z 1017, 12Z 1019, 18Z 1017, 07-13 00Z 1018, 06Z 1018
openweather	temperature	07-10 06Z 26.44, 12Z 29.46, 18Z 32.53, 07-11 00Z 28.77, 06Z 26.31, 12Z 27.3, 18Z 29.96, 07-12 00Z 27.31, 06Z 26.54, 12Z 29.59, 18Z 30.95, 07-13 00Z 27.58, 06Z 26.93
openweather	wind_direction	07-10 06Z 176.2, 12Z 166.2, 18Z 151.4, 07-11 00Z 157.4, 06Z 144.6, 12Z 174.8, 18Z 150.6, 07-12 00Z 154.1, 06Z 169.6, 12Z 168.2, 18Z 138.7, 07-13 00Z 151.6, 06Z 188.4
openweather	wind_speed	07-10 06Z 1.747, 12Z 2.394, 18Z 3.971, 07-11 00Z 2.503, 06Z 1.928, 12Z 2.777, 18Z 3.097, 07-12 00Z 2.538, 06Z 1.705, 12Z 1.188, 18Z 5.163, 07-13 00Z 2.81, 06Z 2.286
purpleair	pm25	07-10 06Z 7.376, 12Z 7.642, 18Z 6.447, 07-11 00Z 8.164, 06Z 8.726, 12Z 7.114, 18Z 6.557, 07-12 00Z 7.236, 06Z 7.474, 12Z 9.754, 18Z 6.124, 07-13 00Z 5.289, 06Z 4.846
sentinel5p	s5p_aer_ai_granule_available	07-10 18Z 1, 07-11 18Z 1, 07-12 18Z 1
sentinel5p	s5p_ch4_granule_available	07-10 18Z 1, 07-11 18Z 1, 07-12 18Z 1
sentinel5p	s5p_co_column	07-10 18Z 0.02624, 07-12 18Z 0.02506
sentinel5p	s5p_co_granule_available	07-10 18Z 1, 07-11 18Z 1, 07-12 18Z 1
sentinel5p	s5p_hcho_column	07-10 18Z 0.0001571, 07-12 18Z 0.0002541
sentinel5p	s5p_hcho_granule_available	07-10 18Z 1, 07-11 18Z 1, 07-12 18Z 1
sentinel5p	s5p_no2_column	07-10 18Z 4.407e-05, 07-12 18Z 3.333e-05
sentinel5p	s5p_no2_granule_available	07-10 18Z 1, 07-11 18Z 1, 07-12 18Z 1
sentinel5p	s5p_o3_granule_available	07-10 18Z 1, 07-11 18Z 1, 07-12 18Z 1
sentinel5p	s5p_so2_column	07-10 18Z 8.954e-06, 07-12 18Z 3.99e-05
sentinel5p	s5p_so2_granule_available	07-10 18Z 1, 07-11 18Z 1, 07-12 18Z 1
tceq	co	07-10 06Z 0.25, 12Z 0.2722, 18Z 0.2278, 07-11 00Z 0.21, 06Z 0.2389, 12Z 0.3167, 18Z 0.2389, 07-12 00Z 0.4, 06Z 0.2222, 12Z 0.2, 18Z 0.2267, 07-13 00Z 0.2889, 06Z 0.1556
tceq	no2	07-10 06Z 5.623, 12Z 6.947, 18Z 5.514, 07-11 00Z 4.932, 06Z 6.486, 12Z 5.291, 18Z 5.577, 07-12 00Z 6.646, 06Z 4.124, 12Z 3.217, 18Z 2.923, 07-13 00Z 4.6, 06Z 2.89
tceq	so2	07-10 06Z -0.1917, 12Z -0.1833, 18Z -0.1889, 07-11 00Z -0.12, 06Z -0.1944, 12Z -0.1833, 18Z 0.5389, 07-12 00Z -0.14, 06Z -0.3706, 12Z -0.3222, 18Z -0.3444, 07-13 00Z -0.3556, 06Z -0.3333

Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

Source	Metric	Values
asos	humidity	T41 (10.667 km) 79.76, EFD (14.102 km) 78.05, LVJ (23.01 km) 78.42, HOU (26.499 km) 78.4, MCJ (39.455 km) 79.16, AXH (45.443 km) 80.28
asos	temperature	T41 (10.667 km) 28.56, EFD (14.102 km) 28.53, LVJ (23.01 km) 28.13, HOU (26.499 km) 28.29, MCJ (39.455 km) 28.17, AXH (45.443 km) 27.33
asos	wind_direction	T41 (10.667 km) 128.3, EFD (14.102 km) 97.41, LVJ (23.01 km) 88.96, HOU (26.499 km) 126.5, MCJ (39.455 km) 133.9, AXH (45.443 km) 53.12
asos	wind_speed	T41 (10.667 km) 2.44, EFD (14.102 km) 2.43, LVJ (23.01 km) 1.547, HOU (26.499 km) 2.311, MCJ (39.455 km) 3.219, AXH (45.443 km) 0.7752
noaa_gfs	gh_500	gfs:29.50,-95.00 (9.357 km) 5926, gfs:29.75,-95.00 (18.624 km) 5925, gfs:29.50,-95.25 (24.49 km) 5925, gfs:29.75,-95.25 (29.288 km) 5925, gfs:30.00,-95.00 (46.387 km) 5926, gfs:29.50,-95.50 (47.768 km) 5925
noaa_gfs	pbl_height	gfs:29.50,-95.00 (9.357 km) 574.2, gfs:29.75,-95.00 (18.624 km) 728.8, gfs:29.50,-95.25 (24.49 km) 488.4, gfs:29.75,-95.25 (29.288 km) 753.6, gfs:30.00,-95.00 (46.387 km) 540.1, gfs:29.50,-95.50 (47.768 km) 586.1
noaa_gfs	precipitable_water	gfs:29.50,-95.00 (9.357 km) 55.32, gfs:29.75,-95.00 (18.624 km) 55.17, gfs:29.50,-95.25 (24.49 km) 55.28, gfs:29.75,-95.25 (29.288 km) 54.89, gfs:30.00,-95.00 (46.387 km) 54.66, gfs:29.50,-95.50 (47.768 km) 55.2
noaa_gfs	surface_pressure	gfs:29.50,-95.00 (9.357 km) 1016, gfs:29.75,-95.00 (18.624 km) 1016, gfs:29.50,-95.25 (24.49 km) 1015, gfs:29.75,-95.25 (29.288 km) 1015, gfs:30.00,-95.00 (46.387 km) 1014, gfs:29.50,-95.50 (47.768 km) 1014
noaa_gfs	t_850	gfs:29.50,-95.00 (9.357 km) 18.73, gfs:29.75,-95.00 (18.624 km) 18.71, gfs:29.50,-95.25 (24.49 km) 18.69, gfs:29.75,-95.25 (29.288 km) 18.65, gfs:30.00,-95.00 (46.387 km) 18.77, gfs:29.50,-95.50 (47.768 km) 18.73
noaa_gfs	u_10m	gfs:29.50,-95.00 (9.357 km) -1.139, gfs:29.75,-95.00 (18.624 km) -1.127, gfs:29.50,-95.25 (24.49 km) -1.29, gfs:29.75,-95.25 (29.288 km) -0.8276, gfs:30.00,-95.00 (46.387 km) -1.168, gfs:29.50,-95.50 (47.768 km) -1.669
noaa_gfs	v_10m	gfs:29.50,-95.00 (9.357 km) 2.741, gfs:29.75,-95.00 (18.624 km) 2.19, gfs:29.50,-95.25 (24.49 km) 2.099, gfs:29.75,-95.25 (29.288 km) 2.114, gfs:30.00,-95.00 (46.387 km) 1.663, gfs:29.50,-95.50 (47.768 km) 2.09
openaq	ozone	5077791 (0.008 km) 0.01424, 332 (14.586 km) 0.02046, 2800 (21.051 km) 0.01662, 320 (24.786 km) 0.01374, 3214 (26.617 km) 0.01486, 339 (26.892 km) 0.01748, 2039 (28.563 km) 0.01642, 3378 (28.774 km) 0.01213, +7 more
openaq	pm10	271 (14.586 km) 26.83, 15852419 (35.786 km) 33.06
openaq	pm25	14404954 (2.969 km) 2.951, 13955815 (13.307 km) 5.233, 268 (14.586 km) 10.42, 11638043 (23.95 km) 2.116, 14157975 (24.789 km) 9.101, 15539682 (28.776 km) 6.591, 8967123 (28.83 km) 7.631, 8967479 (28.83 km) 7.002, +58 more
openweather	cloud_cover	grid:east (21.774 km) 66.57, grid:south (34.665 km) 68.83, grid:center (39.496 km) 64.76
openweather	humidity	grid:east (21.774 km) 83.97, grid:south (34.665 km) 80.17, grid:center (39.496 km) 80.86
openweather	precipitation	grid:east (21.774 km) 0.1272, grid:south (34.665 km) 0.1382, grid:center (39.496 km) 0.08958
openweather	pressure	grid:east (21.774 km) 1017, grid:south (34.665 km) 1017, grid:center (39.496 km) 1017
openweather	temperature	grid:east (21.774 km) 28.43, grid:south (34.665 km) 28.38, grid:center (39.496 km) 28.88
openweather	wind_direction	grid:east (21.774 km) 169.2, grid:south (34.665 km) 159.6, grid:center (39.496 km) 147.8
openweather	wind_speed	grid:east (21.774 km) 3.071, grid:south (34.665 km) 3.01, grid:center (39.496 km) 1.913
purpleair	pm25	208397 (2.117 km) 2.9, 2386 (8.161 km) 0.1589, 3298 (10.11 km) 6.546, 247283 (12.095 km) 7.051, 268165 (12.925 km) 7.218, 222815 (13.892 km) 6.174, 222593 (15.865 km) 6.754, 128007 (15.888 km) 1.529, +54 more
tceq	co	482011039 (14.566 km) 0.1656, 482011035 (28.771 km) 0.1302, 482011052 (44.211 km) 0.4469
tceq	no2	482011050 (0.0 km) 1.914, 482011039 (14.566 km) 3.373, 482011015 (20.507 km) 4.784, 482010026 (26.633 km) 3.128, 482011035 (28.771 km) 7.498, 482010416 (29.325 km) 5.295, 480391004 (37.123 km) 3.598, 482011052 (44.211 km) 11.31, +3 more
tceq	so2	482011039 (14.566 km) 0.2081, 482011035 (28.771 km) -0.3286, 482010051 (44.59 km) -0.3921


Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly
asos (6)
openaq (83)
purpleair (62)
tceq (12)
anomaly
40 km
30 km
20 km
10 km
29.9
29.8
29.7
29.6
29.5
29.4
Latitude (degrees north)
−95.5
−95.4
−95.3
−95.2
−95.1
−95.0
Longitude (degrees east)


Event-nearest wind vectors from stored values (arrow points downwind)
asos
longest arrow = 5.66 m/s
29.70
29.65
29.60
29.55
29.50
−95.5
−95.4
−95.3
−95.2
−95.1
−95.0
noaa_gfs
longest arrow = 3.24 m/s
30.0
29.9
29.8
29.7
29.6
29.5
−95.5
−95.4
−95.3
−95.2
−95.1
−95.0
openweather
longest arrow = 3.49 m/s
29.75
29.70
29.65
29.60
29.55
−95.3
−95.2
−95.1
−95.0

Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). The labeling guide that comes with this packet has the definitions and marking rules. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Claim 1 of 36

The combination of high temperatures and stagnant air patterns led to a buildup of pollutants in the Houston area, causing the observed anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 2 of 36

The moderate temperature and low wind speeds contributed to the stagnation of air pollutants, exacerbating the air quality anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 3 of 36

The air quality anomaly is related to PM2.5 levels.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 4 of 36

Boundary layer heights around 317 meters to 813 meters along with low surface wind speeds around 2.6 to 4.1 m/s limited both vertical mixing and horizontal dispersion during the afternoon period.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 5 of 36

At 2026-07-11T19:55:00+00:00, the surface weather station 10.667 km from the anomaly site recorded a temperature of 30.0 degC and wind speed of 4.12 m/s.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 6 of 36

The pollutant mix near 2026-07-11T20:00:00+00:00 is most consistent with a fresh local NOx plume because NO2 was elevated while nearby CO was about 0.2 ppm, PM2.5 was about 2.2 to 3.3 ug/m3, PM10 was about 23 ug/m3, and ozone was about 0.033 ppm rather than showing a broad multi-pollutant regional buildup.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 7 of 36

The meteorology near 2026-07-11T20:00:00+00:00 was compatible with a concentrated local plume reaching the monitor because surface winds were moderate at about 3.4 to 4.1 m/s and the modeled planetary boundary layer height was only about 317 m at 18 UTC, which would allow limited dilution of a nearby emission plume.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 8 of 36

The anomaly occurred on July 11th, around 20:01:54 UTC, and was located approximately 21.774 km east and 34.665 km south of the query point.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 9 of 36

A severe nitrogen dioxide anomaly was recorded at location (29.583, -95.016) on 2026-07-11T20:00:00+00:00, reaching a value of 9.7 ppb against an expected baseline of 1.23 ppb with a z-score of 6.17.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 10 of 36

A nearby combustion source plume is a leading candidate cause because the anomalous TCEQ NO2 monitor reported 9.7 ppb at 2026-07-11T20:00:00+00:00 versus an expected 1.23 ppb while nearby ozone at the same time was only moderate at about 0.033 ppm, a pattern consistent with fresh NOx influence rather than a broad aged ozone episode.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 11 of 36

Local industrial and traffic combustion emissions from the nearby Houston Ship Channel and Bayport industrial complexes released elevated precursor concentrations, causing the localized NO2 spike recorded by the TCEQ monitor.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 12 of 36

The anomaly is more consistent with a localized NOx event than with a regional smoke or dust intrusion because particulate matter remained low near the anomaly and no contemporaneous regional combustion enhancement is evident in the available carbon monoxide context.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 13 of 36

The planetary boundary layer height dropped to around 317 meters near the anomaly location at 18:00 UTC on July 11, 2026, trapping local surface emissions.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 14 of 36

At the exact time of the nitrogen dioxide anomaly, an ozone measurement of 0.033 ppm was recorded 0.008 km from the anomaly location.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 15 of 36

Meteorology near the anomaly showed only moderate ventilation rather than strong stagnation, with ASOS wind speed near 4.12 m/s at 19:55 UTC and GFS planetary boundary layer height near 317 m at 18:00 UTC, which supports partial trapping of a nearby plume but contradicts an explanation requiring extreme calm or a deeply stagnant boundary layer.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 16 of 36

The tceq NO2 anomaly at 29.583, -95.016 had an expected value of 1.2291970802919707 ppb and a z-score of 6.169790309830889, which was classified as severe.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 17 of 36

The co-pollutant pattern supports a fresh NOx-rich plume and contradicts a broad combustion or regional smog event because NO2 was elevated while nearby ozone was only 0.033 ppm, TCEQ CO was 0.2 ppm, TCEQ SO2 was 0.7 ppb, and nearby PM2.5 remained low at 2.2 ug/m3 in openaq and 3.3 ug/m3 in purpleair at the anomaly time.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 18 of 36

Boundary-layer and surface-weather conditions likely contributed to the NO2 enhancement by limiting vertical dilution relative to a fully mixed afternoon, because gfs placed planetary boundary layer height near 317 m at 18:00 UTC and nearby surface observations showed warm, humid conditions with only moderate winds around 4.1 m/s close to the anomaly time.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 19 of 36

The nearest openaq ozone observation to 29.583, -95.016 at 2026-07-11T20:00:00+00:00 was 0.033 ppm at a sensor 0.008 km from the anomaly location.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 20 of 36

Heavy cloud cover exceeding 90 percent reduced solar irradiance, slowing down the photolytic conversion of NO2 into ozone and allowing NO2 to accumulate near the surface.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 21 of 36

The temperature was unusually high on July 11th, leading to increased emissions from industrial sources and vehicles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 22 of 36

The air quality anomaly in Houston, Texas on July 11th was caused by a combination of factors including high temperatures, low wind speeds, and nearby industrial activities.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 23 of 36

A strong wind pattern on July 11th brought pollutants from nearby cities into the Houston area, contributing to the anomaly.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 24 of 36

High cloud cover reaching 98% near 20:00 UTC suppressed NO2 photolysis rates, favoring NO2 accumulation over atmospheric breakdown.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 25 of 36

The PM2.5 levels are elevated at 3.3 ug/m3, which is higher than the mean of 7.2641 ug/m3.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 26 of 36

The PM2.5 levels were significantly higher than the average in the area, indicating a potential health risk to residents.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 27 of 36

On July 11, 2026, at 20:00 UTC, nitrogen dioxide concentrations monitored by TCEQ in Houston reached an anomalous level of 9.7 ppb against an expected baseline of 1.23 ppb.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 28 of 36

Suppressed vertical dispersion associated with a lower afternoon planetary boundary layer height trapped surface emissions near the ground level.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 29 of 36

The TCEQ NO2 monitor at 29.583, -95.016 recorded 9.7 ppb at 2026-07-11T20:00:00+00:00 versus an expected 1.229 ppb, while the same monitor's mean over the context window was only 1.914 ppb, which supports a localized monitor-scale NOx enhancement rather than a routine regional background level.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 30 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 31 of 36

Overcast conditions with cloud cover reaching up to 98 percent diminished solar photolysis during the afternoon of July 11, 2026.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 32 of 36

The tceq NO2 monitor at 29.583, -95.016 recorded 9.7 ppb at 2026-07-11T20:00:00+00:00.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 33 of 36

The air quality anomaly in Houston, Texas is caused by high levels of SO2 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 34 of 36

The TCEQ NO2 monitor at 29.583, -95.016 measured 9.7 ppb at 2026-07-11T20:00:00+00:00 against an expected 1.229 ppb, which identifies a severe positive NO2 anomaly at that site.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 35 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Claim 36 of 36

A transient increase in sulfur dioxide concentrations recorded by nearby TCEQ monitors between 18:00 and 20:00 UTC indicates localized combustion or industrial activity as a primary emission source.

Mark exactly one:	[ ] V	[ ] I	[ ] U

Note:

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.